

ICAD RT-4 · Set 6 — Answer Sheet

Exam 12-10-2026 · 75 Q · 300 marks · 3 hours · +4 correct, -1 wrong, 0 blank

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Physics

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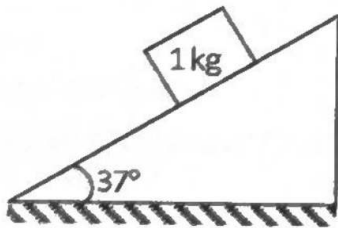
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Bubbles — single correct · Box — write the value · each subject out of 100

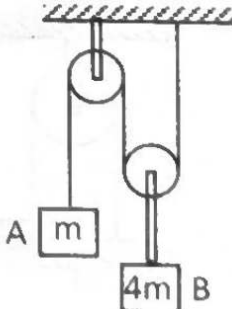
Physics

PART A • Q1-20 • SINGLE CORRECT

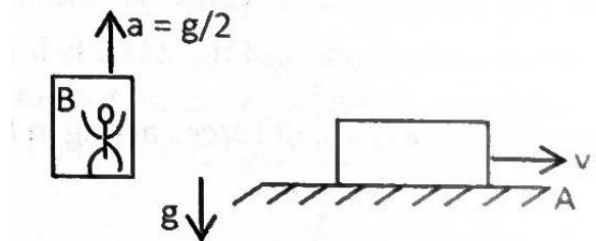
1. P2L5V5Q10 Evaluate: $\int (3x\sqrt{x} + 4\sqrt{x} + 5) dx$ $((6/5)x^{(5/2)} + (8/3)x^{(3/2)} + 5x + C)$. (a) $(6/5)x^{(5/2)} + (8/3)x^{(1/2)} + 5x + C$ (b) $(6/5)x^{(3/2)} + (8/3)x^{(3/2)} + 5x + C$ (c) $(6/5)x^{(5/2)} + (8/3)x^{(3/2)} + 5x + C$ (d) $3x^{(5/2)} + 4x^{(3/2)} + 5x + C$
2. P3L2V4Q3 If $A=B+C$ and the magnitudes of A, B and C are 5, 4 and 3 units respectively, the angle between A and C is: (a) $\cos^{-1}(3/5)$ (b) $\cos^{-1}(4/5)$ (c) $\pi/2$ (d) $\sin^{-1}(3/4)$
3. P3L3L4V6Q8 (Multiple correct) 1 kg block in equilibrium on a 37° incline: friction and normal reaction ($f = 6$ N, $N = 8$ N). (a) The frictional force on the block is zero (b) The frictional force on the block is 6 N (c) The normal reaction on the block is 10 N (d) The normal reaction on the block is 8 N



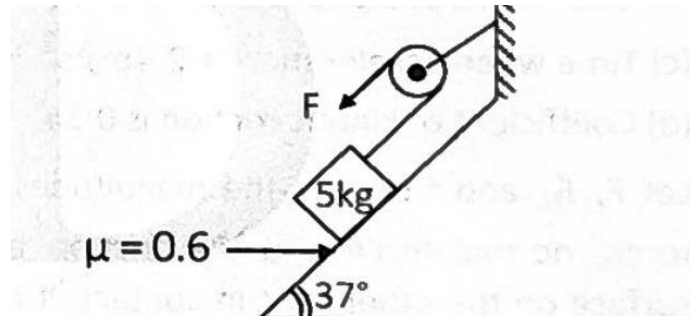
4. P4L3V4Q1 A body is dropped from a height 39.2 m. After it crosses half the distance, the acceleration due to gravity ceases to act. The body will hit the ground with velocity: (a) 19.6 m/s (b) 20 m/s (c) 1.96 m/s (d) 196 m/s
5. P4L3V5Q4 A ball is dropped into a well in which the water level is at a depth h below the top. If the speed of sound is c , the time after which the splash is heard is given by: (a) $h[\sqrt{(2/gh)} + 1/c]$ (b) $h[\sqrt{(2/gh)} - 1/c]$ (c) $h[2/g + 1/c]$ (d) $h[2/g - 1/c]$
6. P4L5V3Q5 A particle's position is $y(t) = 2t^2 + 3t + 4$. Find the average velocity from $t = 0$ to $t = 3$ sec. (a) 3 m/s (b) 6 m/s (c) 9 m/s (d) 12 m/s
7. P4L7V6Q1 A projectile is thrown to have the maximum possible horizontal range equal to 400 m. Taking the point of projection as origin, find the coordinates where the projectile's velocity is minimum: (a) (400,100) (b) (200,100) (c) (400,200) (d) (200,200)
8. P4L9V2Q5 A ball is projected horizontally at 5 m/s from a building 19.6 m high. How long does it take to hit the ground? (a) $\sqrt{2}$ s (b) 3s (c) $\sqrt{3}$ s (d) 2s
9. P6L3V6Q1 Acceleration of the blocks ($A = m$, $B = 4m$; movable-pulley system). (a) $a_A = g/2$, $a_B = g/4$ (b) $a_A = g/6$, $a_B = g/12$ (c) $a_A = g/3$, $a_B = g/6$ (d) $a_A = g/6$, $a_B = g/3$



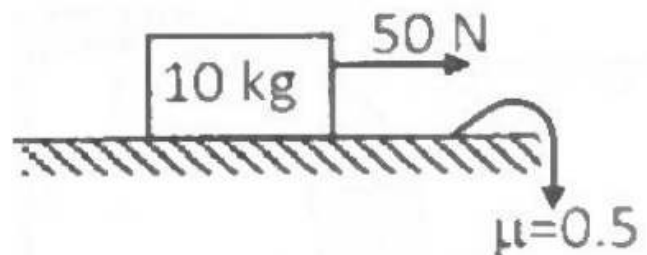
10. P6L6V6Q7 (One or more correct) 10 kg block above a 5 kg block joined by light spring B; 10 kg also attached to a support by spring A. In equilibrium, spring B suddenly cut. Which statements are correct immediately after? (a) Tension in A becomes zero instantly. (b) Both blocks have zero acceleration at $t = 0+$. (c) Upper block's position is unchanged at $t = 0+$, so tension in A is unchanged then. (d) Lower block moves up at $t = 0+$.
11. P7L1V3Q1 Platform A is in free-fall. A block slides with velocity v on surface A, while observer B is in a frame accelerating with $a = g/2$. Compare the value of the friction in the reference frame of B with that in frame A: (a) Remains same (b) Increase (c) Decrease (d) Leftward.



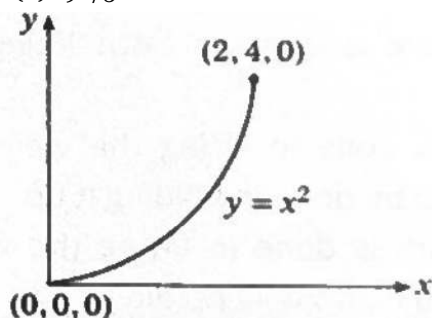
12. P7L2V5Q5 Friction force and its direction on the block by the inclined plane if $F = 10$ N. A 5 kg block rests on a 37° incline with $\mu = 0.6$; a string over a pulley at the top applies a force F up the plane. ($g = 10$ m/s²) (a) 30 N, up the incline (b) 24 N, down the incline (c) 20 N, up the incline (d) none



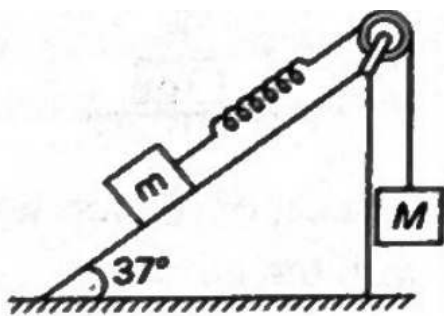
13. P7L3V3Q5 Find the acceleration of the block; initially the block is at rest. A 10 kg block on a rough horizontal floor ($\mu = 0.5$) is pulled by a horizontal force of 50 N. (a) $1/2$ (b) 0 (c) $3/4$ (d) none ($g = 10$ m/s²)



14. P8L1V5Q1 A force $\vec{F} = (3xy - 5z)\hat{j} + 4z\hat{k}$ is applied on a particle. The work done by the force when the particle moves from the point (0, 0, 0) to the point (2, 4, 0) along the curve $y = x^2$ (as shown in the figure) is: (a) $280/5$ J (b) $140/5$ J (c) $232/5$ J (d) $192/5$ J

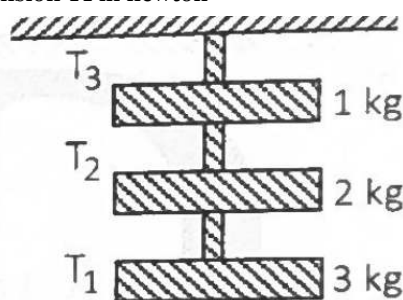


15. P8L2V3Q3 A spring of natural length ℓ is compressed vertically downward against the floor so that its compressed length becomes $\ell/2$. On releasing, the spring attains its natural length. If k is the stiffness constant of spring, then the work done by the spring on the floor is (a) zero (b) $(1/2)k\ell^2$ (c) $(1/2)k(\ell/2)^2$ (d) $k\ell^2$
16. P8L3V6Q4 A block of mass m is attached with a massless spring of force constant k . The block is placed over a rough inclined surface (37°) for which the coefficient of friction is $\mu = 3/4$. The minimum value of M required to move the block up the plane is (neglecting mass of string and pulley and friction in pulley) (a) $2m$ (b) $6m/5$ (c) $4m/5$ (d) $3m/5$

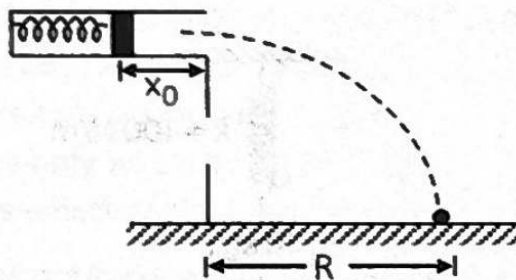


PART B · Q21–25 · NUMERICAL VALUE

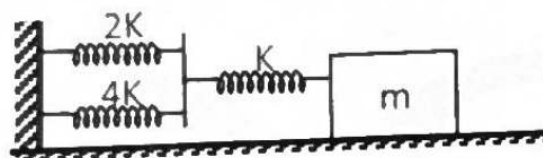
21. P5L2V5Q2 Blocks hanging in series from ceiling: T_3 (top) – 1 kg – T_2 – 2 kg – T_1 – 3 kg (bottom), $g = 10 \text{ ms}^{-2}$; find the tension T_1 in newton



17. P8L6V2Q3 A block of mass 4 kg slides on a horizontal frictionless surface with a speed of 2 m/s. It is brought to rest in compressing a spring in its path. If the force constant of the spring is 400 N/m, by how much the spring will be compressed? (a) 2×10^{-2} m (b) 0.2 m (c) 20 m (d) 200 m
18. P8L8V5Q6 A particle moves with a velocity $(5\hat{i} - 3\hat{j} + 6\hat{k})$ m/s under the influence of a constant force $\vec{F} = (10\hat{i} + 10\hat{j} + 20\hat{k})$ N. Find the instantaneous power applied to the particle. (a) 140 W (b) 200 W (c) 100 W (d) 300 W
19. P1L1V4Q8 Greatest value of $5\cos\theta - 12\sin\theta$. (a) 17 (b) 13 (c) 12 (d) 5
20. P8L6V5Q2 A toy gun whose spring of stiffness k is compressed by a distance x_0 . If the gun is triggered, the shot S of mass m strikes at O. Neglecting friction find the value of R (in m) by using the W–E theorem. (Given: $M = 2$ kg, $K = 400$ N/m, $g = 10 \text{ m/s}^2$, $x_0 = 0.4$ m and $h = 40$ m.) (write the value)



22. P6L6V5SuggQ3 Find k_{eq} : 2K and 4K in parallel, then K in series, block m on a smooth surface



23. P2L6V4Q1 Area enclosed by parabola $y = 3x^2$, the positive x-axis, positive y-axis and the line $y = 12$
24. P3L3L4V3Q2 If $\vec{A} = 4\hat{i} - 3\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$, magnitude of $\vec{A} + \vec{B}$ is $5\sqrt{x}$: find x .
25. P4L11V3Q3 Man holds umbrella at 30° with vertical. Throws umbrella, runs at 10 km/hr; raindrop hits head vertically. Find speed (in kph) of raindrops w.r.t. road.

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L7V3Q3 Most probable speed of a gas ($M=40$) at 127°C . (a) 400 m/s (b) 421 m/s (c) 449 m/s (d) 460 m/s
2. C1L8V3Q3 Real gas $PV=25.8$, $n=1$, $R=0.0821$, $T=300$ K – Z and deviation. (a) $Z = 1.05$, Positive deviation (b) $Z = 1.05$, Negative deviation (c) $Z = 0.95$, Positive deviation (d) $Z = 1.00$, Ideal behavior

3. C2L2V4Q5 EM radiation $\lambda = 20 \text{ nm}$ – frequency and identify region. (a) $1.50 \times 10^{16} \text{ s}^{-1}$ - X-ray (b) $1.50 \times 10^{15} \text{ s}^{-1}$ - UV (c) $3.00 \times 10^{14} \text{ s}^{-1}$ - Visible (d) $1.50 \times 10^{13} \text{ s}^{-1}$ - IR

4. C2L2V6Q9 Assertion: When a black body is heated, it emits radiation only of certain discrete energies and not a continuous range. Reason: According to Planck's hypothesis, energy exchange between matter and radiation occurs in small, indivisible packets called quanta, each having energy equal to $h\nu$. (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true, but R is false. (d) A is false, but R is true.
5. C2L7V3Q1 Find the wavelength for the transition $n=3 \rightarrow 2$. (a) 410 nm (b) 434 nm (c) 486 nm (d) 656 nm
6. C2L9V5Q3 Correct set of four quantum numbers for the valence electron of Rb ($Z=37$). (a) 5, 0, 0, $+1/2$ (b) 5, 1, 0, $+1/2$ (c) 5, 1, 1, $+1/2$ (d) 6, 0, 0, $+1/2$
7. C3L6V3Q2 $2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$: identify limiting reagent for 2 mol Al + 3 mol Cl_2 . (a) Al (b) Cl_2 (c) Both (d) None
8. C3L8V5Q3 120 g urea ($M_w = 60$) in 1000 g water, density 1.15 g mL^{-1} ; molarity? (a) 1.78 M (b) 2.00 M (c) 2.05 M (d) 2.22 M
9. C4L1V4Q6 An element X belongs to the third period of the p-block elements and has 4 electrons in the outermost shell. Name the element. (a) Si (b) P (c) S (d) Al
10. C4L6V4Q2 Among the elements with atomic numbers 9, 12, 16 and 36 which is highly electropositive? (a) Element with atomic number 9 (b) Element with atomic number 12 (c) Element with atomic number 16 (d) Element with atomic number 36
11. C5L1V4Q9 A chemical bond is formed between two atoms when: (a) Forces of repulsion exceed forces of attraction (b) Total energy of the system increases (c) Net result of attraction is greater, causing decrease in energy (d) Electrons stop participating in bonding
12. C5L2V5Q4 How many and what types of bonds are present in NH_4^+ ? (a) Four covalent bonds (b) Three covalent bonds and one ionic bond (c) Four ionic bonds (d) Three covalent bonds and one coordinate bond
13. C5L7V3Q1 Why is a bonding molecular orbital lower in energy than the corresponding atomic orbitals? (a) Due to repulsion between nuclei (b) Due to increased electron density between nuclei (c) Due to formation of nodal plane (d) Due to reduced nuclear charge
14. C5L7V5Q3 During the formation of a molecular orbital from atomic orbitals, the electron density is: (a) Minimum in the nodal plane (b) Maximum in the nodal plane (c) Zero in the nodal plane (d) Zero on the surface of the lobe
15. C5L8V3Q3 In NO_3^- , calculate average bond order using resonance concept. (a) 1.33 (b) 1.5 (c) 2 (d) 1
16. C1L8V4Q2 250 mL A (600 mmHg) + 500 mL B (700 mmHg) into 1.0 L flask holding 50 mL air (760 mmHg) — final total P. (a) 500.0 mmHg (b) 538.0 mmHg (c) 545.0 mmHg (d) 560.0 mmHg
17. C2L10V3Q5 How many electrons are in the $n=4$ shell considering all subshells? (a) 18 (b) 24 (c) 32 (d) 36
18. C4L2V6Q10 An element X has the following electron configuration arranged in Slater groups: $(1s^2)(2s^2 2p^6)(3s^2 3p^6 3d^{10})(4s^2 4p^3)$. Using Slater's rules for an ns/np electron, calculate the effective nuclear charge Z^* felt by a 4p electron. The atomic number of X is $Z = 33$. (a) 5.45 (b) 6.30 (c) 6.85 (d) 7.50
19. C5L5V3Q3 A central atom has 4 bond pairs and 1 lone pair. Find Q and shape. (a) $Q=5$, See-saw (b) $Q=4$, Tetrahedral (c) $Q=5$, Trigonal bipyramidal (d) $Q=6$, Square pyramidal
20. C1L5V5Q9 Diffusion velocities 1:4, masses in mixture 2:3 — mole ratio A:B. (a) 3 : 8 (b) 8 : 3 (c) 1 : 4 (d) 1 : 24

PART B · Q21–25 · NUMERICAL VALUE

21. C1L10V4Q3 Plate 25 cm^2 feels 50 dyn at 5.00 cm/s over 0.25 cm — find η . (a) 0.10 poise (b) 1.00 poise (c) 0.50 poise (d) 2.00 poise
22. C3L5V4Q2 10 grams H_2 can produce how many moles of water?
23. C3L7V4Q2 50 mL of 20% v/v ethanol solution contains how many mL of ethanol?
24. C3L7V6Q1 50 mL of 20% w/w urea solution is mixed with 100 mL of 10% w/w urea solution. Determine % w/w of resultant solution.
25. C3L5V5Q5 3.011×10^{23} molecules of sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) consume x litre of O_2 during metabolism. Calculate $x/44.8$

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M1L2V4Q6 Distance by tip of 12 cm hour hand, 6:00 a.m.–3:15 p.m. (a) $35/2 \pi \text{ cm}$ (b) $18\pi \text{ cm}$ (c) $37/2 \pi \text{ cm}$ (d) $19\pi \text{ cm}$
2. M2L2V4Q4 $(A \cup B) \cap (A' \cap B')$ equals (a) ϕ (b) Ω (c) A (d) B
3. M3L3V3Q5 $\cot(135^\circ) + \tan(360^\circ)$. (a) -1 (b) 0 (c) 1 (d) 2
4. M4L3V5Q9 Domain of $f(x) = \log_2(x+3)/(x^2+3x+2)$. (a) $\mathbb{R} - \{-1, -2\}$ (b) $(-2, \infty)$ (c) $\mathbb{R} - \{-1, -2, -3\}$ (d) $(-3, \infty) - \{-1, -2\}$
5. M5B1V5Q10 If in a $\triangle ABC$, $\tan A + \tan B + \tan C = 6$, then $\cot A \cot B \cot C = ?$ (a) 6 (b) 1 (c) $1/6$ (d) none
6. M5AL3V4Q6 $(1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2$ is equal to: (a) $\tan^2 \alpha + \tan^2 \beta$ (b) $\cos^2 \alpha \cos^2 \beta$ (c) $\sec^2 \alpha \sec^2 \beta$ (d) $\tan^2 \alpha \tan^2 \beta$
7. M5CL3V3Q1 If $\tan(\pi/4 + \theta) + \tan(\pi/4 - \theta) = \lambda \cdot \sec 2\theta$, then $\lambda =$ (a) 3 (b) 4 (c) 1 (d) 2
8. M5CL3V4Q2 $(\sqrt{2} - \sin \alpha - \cos \alpha)/(\sin \alpha - \cos \alpha) =$ (a) $\sec(\alpha/2 - \pi/8)$ (b) $\cos(\pi/8 - \alpha/2)$ (c) $\tan(\alpha/2 - \pi/8)$ (d) $\cot(\alpha/2 - \pi/2)$
9. M6L3V6Q5 For $a > 0$, solve $|x - a| + |x - 3a| = |x - 2a| + a$. (One or more options may be correct.) (a) No solution for $a < 1/2$ (b) Exactly three real solutions for each $a > 0$ (c) One root always at $x = 2a$ (d) Solutions symmetric about $x = 2a$
10. M6L4V5Q4 If $|x - 2| + |x - 3| = 7$, then $x =$ (a) 6 (b) -1 (c) 6, -1 (d) None
11. M7L1V6Q6 A function F is defined on $D = \{x > 0 : x \neq 1\}$ by $F(x) = \log_x(x+1) + \log_x(x/(x+1)) + \log_x(x^2)(x^2-1)$. For how many real $x \in D$ does $F(x) = 1$ hold? (a) 0 (b) 1 (c) 2 (d) 3
12. M7L4V4Q5 If $3^{2x+1} \cdot 4^{x-1} = 36$, then $x =$ (a) $\log_{36} 48$ (b) $\log_{48} 36$ (c) $\log_{24} 12$ (d) $\log_{12} 24$

- 13.** M8AL1V5Q10 Match the following: (P) $a = -18$, $n = 10$, $d = 2$, then a_n of the A.P. ; (Q) a , b and c in A.P., then their arithmetic mean is ; (R) If 2, 4, 6 are in A.P. then 4, 8, 12 will also be an ; (S) If $a_n = 9 - 5n$ of an A.P. then a_{10} will be ; (T) If $d = -2$, $n = 5$ and $a_n = 0$ in the A.P. then a is — with Column-II (a) $(a+c)/2$, (b) 0, (c) -41 , (d) 8, (e) A.P. Options: (a) $P \rightarrow b$, $Q \rightarrow a$, $R \rightarrow e$, $S \rightarrow c$, $T \rightarrow d$ (b) $P \rightarrow a$, $Q \rightarrow c$, $R \rightarrow e$, $S \rightarrow b$, $T \rightarrow d$ (c) $P \rightarrow c$, $Q \rightarrow d$, $R \rightarrow b$, $S \rightarrow a$, $T \rightarrow e$ (d) $P \rightarrow e$, $Q \rightarrow a$, $R \rightarrow b$, $S \rightarrow d$, $T \rightarrow c$
- 14.** M8AL2V3Q5 If the sum of n terms of an A.P. is $2n^2 + 5n$, then its n th term is (a) $4n - 3$ (b) $3n - 4$ (c) $4n + 3$ (d) $3n + 4$
- 15.** M8AL3V4Q7 The ratio of the sums of first n even numbers and n odd numbers will be (a) $1 : n$ (b) $(n + 1) : 1$ (c) $(n + 1) : n$ (d) $(n - 1) : 1$
- 16.** M8BL2V5Q4 The sum of the series $6 + 66 + 666 + \dots$ up to ' n ' terms is (a) $(10n + 1 - 9n + 10)/81$ (b) $2(10n + 1 - 9n - 10)/27$ (c) $2(10n - 9n - 10)/27$ (d) None of these
- 17.** M8CL1V5Q5 If $a^x = b^y = c^z$ and a, b, c are in G.P., then x, y, z are in (a) A.P. (b) G.P. (c) H.P. (d) None
- 18.** M2L3V5UggQ9 140 students (1–140): evens $\rightarrow$ Maths, $\div 3 \rightarrow$ Physics, $\div 5 \rightarrow$ Chemistry $\rightarrow$ opted none. (a) 102 (b) 42 (c) 1 (d) 38
- 19.** M8BL1V4Q3 If 5th, 8th and 11th term of a G.P. are 9, λ and 25 then λ is ($\lambda > 0$) (write the value)
- 20.** M8BL2V3Q3 Find sum of the infinite G.P. 3, 1, $1/3$, $1/9$, (write the value)

PART B · Q21–25 · NUMERICAL VALUE

- 21.** M8AL3V4Q1 A man saved Rs. 16500 in ten years. In each year after the first he save Rs. 100 more than he did in the preceding year. How much did he save in the first year?
- 22.** M4L1V2Q4 Non-empty subsets of $\{1, 2, 3, 4\}$
- 23.** M6L4V4Q8 If solution set of $(x^2 - 5x + 6) < 0$ is $x \in (a, b)$ then $a + b$ is _____
- 24.** M1L2V4Q1 Diameter 66 cm, chord 33 cm; minor arc = $K\pi$ cm, find K
- 25.** M5CL1V5Q7 If $\cos 4x = 1 - k \sin^2 x \cos^2 x$, then k is: